BEAUVILLE-LASZLO GLUEING REALIZES EVERY QUASI-AFFINE TRIPLE BY A SEPARATED SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2757-2764 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2766-2844, the complete author proof (77 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_spaces-pushouts.json (pb=2899, stmt 2757-2764, proof 2766-2844).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04S6: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2844–2844. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9N: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2470–2474. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2547–2555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9R: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2582–2589. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

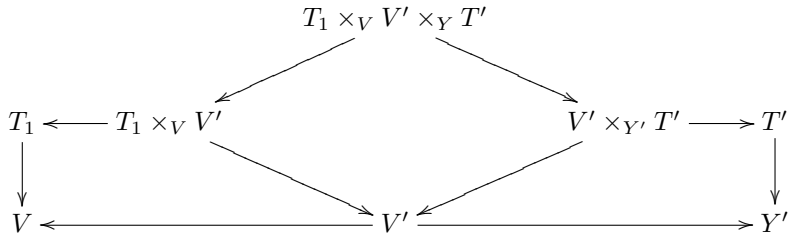
Source reference. Original lemma 0F82: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2844–2844. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0420: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2844–2844. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(R \to R', f)$ be a glueing pair, see above. Any object (V, V', Y') of Spaces$(U \leftarrow U' \to X')$ with V, V', Y' quasi-affine is isomorphic to the image under the functor (0F9N) of a separated algebraic space Y over X.*

Proof. Choose n', $T' \to Y'$ and n_1, $T_1 \to V$ as in Properties, Lemma 0F82. Picture



Observe that $T_1 \times_V V'$ and $V' \times_{Y'} T'$ are affine (namely the morphisms $V' \to V$ and $V' \to Y'$ are affine as base changes of the affine morphisms $U' \to U$ and $U' \to X'$). By construction we see that

$$\mathbf{A}^{n'}_{T_1 \times_V V'} \cong T_1 \times_V V' \times_{Y'} T' \cong \mathbf{A}^{n_1}_{V' \times_{Y'} T'}$$

In other words, the affine schemes $\mathbf{A}^{n'}_{T_1}$ and $\mathbf{A}^{n_1}_{T'}$ are part of a triple making an affine object of *Spaces*$(U \leftarrow U' \to X')$. By Lemma 0F9Q there exists a morphism of affine schemes $T \to X$ and isomorphisms $U \times_X T \cong \mathbf{A}^{n'}_{T_1}$ and $X' \times_X T \cong \mathbf{A}^{n_1}_{T'}$ compatible with the isomorphisms displayed above. These isomorphisms produce morphisms

$$U \times_X T \longrightarrow V \quad \text{and} \quad X' \times_X T \longrightarrow Y'$$

satisfying the property of Properties, Lemma 0F82 with $n = n' + n_1$ and moreover define a morphism from the triple $(U \times_X T, U' \times_X T, X' \times_X T)$ to our triple (V, V', Y') in the category *Spaces*$(U \leftarrow U' \to X')$.

By Lemma 0F9Q there is an affine scheme W whose image in *Spaces*$(U \leftarrow U' \to X')$ is isomorphic to the triple

$$((U \times_X T) \times_V (U \times_X T), (U' \times_X T) \times_{V'} (U' \times_X T), (X' \times_X T) \times_{Y'} (X' \times_X T))$$

By fully faithfulness of this construction, we obtain two maps $p_0, p_1 : W \to T$ whose base changes to U, U', X' are the projection morphisms. By Lemma 0F9R the morphisms p_0, p_1 are flat and of finite presentation and the morphism $(p_0, p_1) : W \to T \times_X T$ is a closed immersion. In fact, $W \to T \times_X T$ is an equivalence relation: by the lemmas used above we may check symmetry, reflexivity, and transitivity after base change to U and X', where these are obvious (details omitted). Thus the quotient sheaf

$$Y = T/W$$

is an algebraic space for example by Bootstrap, Theorem 04S6. Since it is clear that Y/X is sent to the triple (V, V', Y'). The base change of the diagonal $\Delta : Y \to Y \times_X Y$ by the quasi-compact surjective flat morphism $T \times_X T \to Y \times_X Y$ is the closed immersion $W \to T \times_X T$. Thus Δ is a closed immersion by Descent on Spaces, Lemma 0420. Thus the algebraic space Y is separated and the proof is complete. □